

**e-Repository of Researches for Camarines Norte Lowland
Rainfed Research Station (CNLRRS)**

A Capstone Project Presented to the Faculty of the
College of Computing and Multimedia Studies
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In Partial Fulfillment of the Requirements
For the Degree in Bachelor of Science
in Information System

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ABSTRACT

Title: e-Repository of Researches for
Camarines Norte Lowland Rainfed
Research Station (CNLRRS)

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This study explores the development of a web-based electronic repository system to improve access, organization, and preservation of research outputs from the Camarines Norte Lowland Rainfed Research Station (CNLRRS). To address challenges in traditional document handling—such as limited accessibility, misplacement of files, and time-consuming retrieval—the researchers designed the CNLRRS e-Repository as a centralized digital platform for storing and managing research documents. Using the Agile Methodology, iterative development cycles allowed for continuous evaluation and refinement of the system based on user feedback. Key features such as document upload, search filters, automated metadata management, and role-based access were integrated to enhance usability and efficiency. An evaluation conducted among CNLRRS researchers and staff showed a high

usability rating with an overall weighted mean score of 4.53. Based on the findings, the study concludes that systems like the CNLRRS e-Repository are effective tools for improving data accessibility, research management, and long-term knowledge preservation within agricultural research institutions.

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The Researchers

DEDICATION

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The Researchers

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Chapter 1

INTRODUCTION

Context of the Study

The Camarines Norte Lowland Rainfed Research Station (CNLRRS) is a subsidiary office of the Department of Agriculture, Region 5, located in Barangay Calasgasan, Daet, Camarines Norte. The CNLRRS conducts major research in crops in the Bicol region, specifically Queen Pineapple, the main crop produced in Camarines Norte. They have already developed a new strain of Queen pineapple referred to as MD2, which represents a major milestone for the Office. The current distribution of research data at CNLRRS makes it challenging to access, discuss, and expand on previous findings. In pineapple farming, the lack of a centralized system hinders innovation and leads to operational inefficiencies. In the digital age, e-libraries have emerged as valuable tools for enhancing national food security by providing access to a vast array of agricultural knowledge, research findings, and best practices (Kamal et al., 2025).

Digital repository systems are effective at managing corporate knowledge. By safely storing and granting access to research outputs, centralized web-based

platforms effectively address information dissemination issues, according to information systems. Digital platforms enhance information sharing and collaboration among stakeholders, transforming research into useful applications (Abdullah, 2022).

This platform will serve as a centralized digital library for CNLRRS's pineapple research, facilitating the sharing, preserving, and storage of important research on crop varieties, rain-fed farming, and pest control. Enhancing access to this specialized knowledge will promote cooperation and strengthen the station's research programs' overall impact. The entire pineapple farming community will benefit from data-driven decisions that improve production and guide agricultural policies by bridging the gap between research and real-world farming applications. Library outreach services can leverage modern information technology to bridge the gap between distant unserved library users and library facilities for speedy and real-time information delivery and exchange (Johnson & Hill, 2022).

Research Objectives

The main objective of this study is to design and develop an e-Repository of Researches for CNLRRS.

Specifically, it aims to:

1. Determine the information requirements needed to develop the e-Repository of Researches for CNLRRS;
2. Determine the features and functionalities of the e-Repository of Researches for CNLRRS; and
3. Determine the level of effectiveness of the developed system in terms of:
 - a) Functionality;
 - b) Performance Efficiency;
 - c) Security

Research Questions

The study was conducted to address the overall objective of designing and developing an E-Repository of Researches for Camarines Norte Lowland Rainfed Research Station (CNLRRS). Specifically, this answers the following:

1. What information is required to develop the e-Repository of Researches for CNLRRS?
2. What specific system features of the e-Repository of Researches for CNLRRS; and
3. What is the level of effectiveness of the developed system in terms of:
 - a) Functionality;
 - b) Performance Efficiency;

c) Security?

Scope and Limitations

The aim of this research is to help manage and maintain agricultural research outputs conducted in the Camarines Norte Lowland Rainfed Research Station (CNLRRS) through researching, conceptualizing, designing, and developing an e-Repository of Researches. The extent of this research is the creation of a web-based centralized repository system that enables researchers, personnel, and administrators to effectively store, sort, and retrieve research data and publications. The system comprises core components such as the e-Library Module, a database repository for finalized research works, publications, and reference materials on pineapple cultivation and other farm-related studies.

The system has a Data Management System to keep track of research data, a User Management Module that provides data protection through regulated user access, and a Search and Retrieval Function to allow for easy access and traversal of stored information. Additionally, the system provides report generation features that make it easy to produce research abstracts and catalogs of outputs. With web-browser access across desktop and

mobile devices, the platform enhances efficiency, organization, and online preservation of CNLRRS research outputs, making it a tool for institutional record-keeping and research visibility.

A key limitation of this study is that the system requires a stable internet connection, which may be difficult for users in remote areas with limited access to WiFi. Additionally, the system is not equipped with real-time plagiarism checkers, which may limit its usefulness and functionality. While the platform is beneficial for data management and research, it lacks a wide scope of content. It does not include features for managing raw research data, collaborative workspaces for ongoing projects, or tools for real-time data analysis and quality control. The system's value depends on consistent and accurate data entry from its users. Training and orientation may also be required for users to fully understand and utilize the system, which may affect satisfaction and ease of learning. Lastly, user adoption depends on the potential needs within the agency, which will require comprehensive training and change management to encourage researchers to transition from traditional archiving methods to the digital platform.

Significance of the Study

The result of the study will be beneficial for the following entities:

CNLRRS Management. The repository system will enhance knowledge preservation and accessibility by centralizing and preserving the institution's research outputs. Systematically documenting studies will improve trust in institutions and facilitate more effective research tracking.

Researchers at CNLRRS. Simplified access to both historical and modern research will help researchers by simplifying literature reviews and avoiding unnecessary duplication. The system will enhance research consistency and create opportunities for collaboration.

Agriculture Department. The platform will enhance data-driven decision-making and program development within the pineapple industry by providing stakeholders structured access to reliable research.

Academic Community. The system will serve as a significant digital resource for students and researchers studying agricultural science, giving accurate reference materials and displaying effective ICT (Information and Communications Technology) uses in research management.

Agricultural Extension Workers. These experts will have efficient access to verified research results that can be converted into actionable guidance for pineapple farmers.

Respondents of CNLRRS:

Contributors from the Camarines Norte Lowland Rainfed Research Station (CNLRRS) made valuable inputs that supported the enhancement of the proposed E-Repository of Researches. Engr. Eula D. Rada, serving as the Research Unit Head, recommended incorporating multiple export formats to enhance data accessibility and usability. Mr. Jude Mike C. Castro suggested providing standardized template for research publication, and Ms. Erosa Tolosa proposed developing a system monitoring dashboard to assist administrators in overseeing platform activities. Furthermore, Ms. Jyne E. Ramos emphasized the need for progress indicators for lengthy upload processes, and Ms. Flor Angielyn B. Campita recommended implementing customizable search result display. Mr. Segundo F. Sorita highlighted the importance of including a duplicate identification system for efficient file management, and Ms. Josephine A. Manlangit advised integrating a plagiarism checker to uphold research integrity. In addition, Ms. Vera S. Canet proposed adding

"cited by" functionality to monitor research impact; Mr. Edton B. Ecleo suggested enhancing the search bar with advanced filter feature, and Mr. Jhon-Jhon D. Galero recommended incorporating system usage trend analysis reports. Collectively, their insights and recommendations provided a valuable foundation.

Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This section of the paper examined relevant information from diverse sources and viewpoints that either support or are pertinent to the current topic. It contained all significant writings, citations, assertions, and research by different experts on the study's topic.

Information Requirement Needed to Develop an E-Repository of Researches for CNLRRS

Library outreach services involve providing and delivering information resources to library users outside the library building, specifically involving bringing library and information services to the doorstep of patrons who are unable to visit the library for some reason. There are several examples of library outreach services in existence, such as bookmobile, library on the go, Info bus, and library elderly outreach project. It is also envisaged that library outreach services can utilize present-day information technology to bridge the gap between distant unserved library users and library facilities for speedy and real-time information delivery and exchange (Johnson & Hill, 2022).

According to Benson et al. (2021), library and information service is an aspect of social service rendered to society at different degrees and levels to ensure that members of society at any time have access to timely and accurate information relevant to their information needs.

The study revealed that the e-library is very helpful in their studies and they are getting benefits from it. It has been suggested that library staff should focus on solving problems such as lack of computers and a lack of access to e-resources outside the library (Iqbal et al., 2022).

The study concluded that while agricultural colleges in Northwest Nigeria possess diverse and relevant information resources, further efforts are needed to support and expand these collections. Recommendations for enhancing access and preservation of these resources are provided (Magaji et al., 2024).

There has been a substantial increase in research publications on various subjects, particularly in the past ten years. Factors determining the popularity of articles in the e-library sector include keywords searched in Scopus, the number of citations, and the publisher's name. The insights gained from this study,

along with the discussions on each issue, will guide researchers, academicians, journal editors, and practitioners in directing their research efforts (Verma et al., 2025).

Electronic library (E-library) is a form of computer-mediated system that uses electronic media, such as Web/internet devices and distributes resources to improve the quality of teaching and learning. Students' use of e-library for learning is essential, and as such, the government has invested hugely into its subscription for several university libraries in Ghana (Okyere-Kwakye & Md Nor, 2022).

The main purpose of using e-library services is research and the benefits of using e-library services include accessing information 24/7, finding relevant information, and performing well in academic endeavours. The majority of postgraduate students were satisfied with the usefulness of e-library services while also perceiving their skills for accessing e-library services as not satisfactory (Abdullah, 2022).

The growth of the internet has emerged as a powerful tool and changed the methods of research, storage, retrieval and communication of scholarly information in the higher education system. The use of different

e-resources enabled the users to retrieve relevant information. With the growing importance of e-resources in education and research, agricultural educational institutes are stepping ahead to keep pace with the latest advances in information technologies (Meenambigai & Keerthana, 2022).

E-library is a medium through which electronic materials and/or resources such as online journals, e-books, academic databases, web directory, etc. are accessed using internet-enabled devices (Chime & Ekwueme, 2023) consumed by students, teachers, and other researchers. Chime and Ekwueme (2023) averred that these materials provided by the e-library are enhancers of research and academic work.

Libraries play a pivotal role in the provision of this information since they are credible sources of information. This article models e-library services for public policy and sustainable development in Kenya (Owiti & Kwanya, 2023).

According to Kehinde & Adeyeye (2023), this study investigated the influence of library user education on use of information resources by students of Federal Colleges of Agriculture Ibadan, Oyo State. It was recommended that Librarians have to adopt library

orientation, online instruction and course-related instruction as a form of library user education to promote massive use of their information resources.

The use of technology in the form of e-library applications, especially e-books, is an important element in increasing the digital literacy among educators and students. This research aims to determine the use of e-books in e-library applications in increasing digital literacy of users. Observation, interviews, and documentation as data collection methods. The use of e-books in e-library applications also contributes to increasing ease of access to information and the level of digital literacy for users (Farika et al., 2024).

Electronic libraries, or virtual libraries, which are popularly known as e-libraries, began to emerge in the 1970s. Recently, it has risen to the internet stage, with it, the web development and web-based resources and access tools, which include databases. It was recommended that there should be a comprehensive education programme for users of the e-library in order to prepare them with the prerequisite knowledge and skills on information technology (Abdullahi et al., 2024).

Drafting of the manuals for the effective running of the library, such as recommendations on the kind of

infrastructure in terms of hardware, software, and networking required to establish the library systems, a user manual for recommendations for additional training resources for the library, and a user training manual for e-library systems (Phiri & Mwale, 2022).

Camacho et al. (2022), emphasize that centralized digital resources contribute significantly to agro-industrial innovation. In their study on pineapple biomass utilization in Costa Rica, they demonstrated that accessible e-library data supported research into converting pineapple waste into biomedical materials. Their findings highlight how digital libraries not only preserve valuable agricultural knowledge but also open opportunities for interdisciplinary applications of pineapple-related research.

The analysis of related studies and literature on information requirements revealed the fundamental data elements, user needs, and system specifications necessary for developing an effective e-repository. These include accurate data categorization, user access levels, metadata standards, and information-sharing protocols. Understanding these requirements guided the formulation of the system design. And ensured that the proposed E-Repository of Researches for the Camarines Norte

Lowland Rainfed Research Station (CNLRRS) aligns with institutional goals and user expectations. The relevance of this study lies in its potential to improve research data management, promote accessibility, and enhance the overall efficiency of research operations within CNLRRS.

Features of the E-Repository of Researches for Camarines Norte Lowland Rainfed Research Station (CNLRRS)

Electronic libraries are systems that combine the machinery of digital computing, storage, and communication, in which the content, and software needed to reproduce and emulate learning extend that of collecting, cataloguing, finding and disseminating information offered by traditional libraries (Li & Furht, 2014). Electronic libraries can also be described as a system providing a community of users with coherent access to a large, organized repository of information and knowledge (Akinselure & Idubor, 2021).

The findings reveal differences in research trends within various elements of electronic libraries. There has been a substantial increase in research publications on various subjects, particularly in the past ten years. Factors determining the popularity of articles in the e-library sector include keywords searched in Scopus, the

number of citations, and the publisher's name (Verma et al., 2025).

Integrating digital library systems like SLiMS creates new pathways for literacy development by providing interactive platforms that combine traditional reading with digital information skills. Through the proliferation of e-books and online resources accessible via computers, smartphones, and tablets, users now benefit from enhanced portability, space efficiency, and environmental sustainability (Humaidi et al., 2025).

In the present study, the researcher collected data through a well-structured questionnaire based on the demographic profile, library membership details, library resources, services and infrastructure facilities, to identify the utilization of public library services in Madurai District. The recent developments in the field of e-library should be made known to the public and have become essentially imperative in this dynamic branch of library services (Thanuskodi, 2025).

In the digital age, e-libraries have emerged as valuable tools for enhancing national food security by providing access to a vast array of agricultural knowledge, research findings, and best practices. These digital repositories enable farmers, policymakers,

researchers, and other stakeholders to access real-time data on improved farming techniques, pest control, soil management, and climate adaptation strategies (Kamal et al., 2025).

The library is known as a knowledge resource centre in an academic institution, where its users gather information from various information sources based on their information needs. Library websites can offer far more services than a traditional physical library in the digital context (Kiran et al., 2021).

The e-library, on the other hand, offers access to a wide range of digital resources, which is increasingly important for accessing up-to-date information in agricultural practices (Ojo & Adekoya, 2022). The similar scores suggest that both in-person and digital engagements are valued by users (Olusanya et al., 2025).

Study suggests that certain interface characteristics, such as terminology, screen design, and navigation, indirectly affect perceived usefulness by influencing perceived ease of use in e-library systems. These findings underscore the significance of e-library system characteristics, particularly system quality. In addition to enhancing accessibility, a web-based e-library platform facilitates the centralization of

digital resources, making it easier for librarians and administrators to manage and maintain a vast collection of materials (Bareer & Mohammadi, 2023).

According to Mondal (2021), digital libraries give acceptable opportunities for learners, attributable to as well as up-to-date data. aggregation transmission resources it doable for remote learning. Digital libraries by storing and making possible the retrieval of academic resources from laptop networks provided valuable aid aid to electronic and networked learning. While there is no existence of organized and maintained electronic resources by digital libraries, accessibility to up-to-date eLearning resources isn't fully crammed.

In this digitalized world, e-resources have become the most functional tool for higher educational institutes. It is advantageous in improving search speed, provides more access points, the ability to access information that would otherwise be unavailable, and an increasing amount of information is in electronic form. Whereas the old library system consumes a lot of valuable time for the scholars (Prajapati et al., 2022).

Integrating electronic resources into library services can have a number of advantages, such as boosting efficiency and offering better user services,

expanding access to resources and fostering new research, assisting in conservation and preservation and enhancing the value of the parent organization's collection, supporting e-learning and online research, and supplanting traditional print resources with multiple digital formats. Compared to print publications, electronic information sources, according to Bhardway & Shama (2022), are more adaptable. They offer superior searching capabilities by utilizing full text or keyword indexing (Olanusi & Edward 2022).

In this digital age, library resources and services have experienced a shift from the traditional library concept, which was mainly physical, to comprise both physical and digital (Volkova & Shrayberg, 2020). The concept of a digital library allows users much-needed access to information with ease, without time or geographical limits. That is, a library user can access the library remotely with or without any physical assistance for enhanced learning and research (Adebimpe et al., 2021).

Key features include open study areas without designated private study spaces, an e-library that ensures both on-site and remote access, conference rooms for group activities, and office spaces for

administrative operations. Additional facilities include specialized sections for phone calls, relaxation, dining, and charging stations, making the library a multifunctional hub for academic and personal use (Akinola & Amadhe, 2025).

Everyone found the conventional method of reading study materials to be unpleasant and time-consuming, and the traditional library only contained a small number of copies of each subject. That is the reason in the era of new advanced technology, traditional libraries are gradually moving towards an e-library (Sonkar, 2014). E-resources play a crucial role in enhancing education for the majority of colleges, institutions, and schools with a higher level of education (Yadav, 2023).

The aim of the e-Library is to disseminate information, share good practices, and promote current awareness to assist in underpinning written work, add value and depth to our students' study experience, and support the continuing professional development of all those working and aspiring to work in the built environment (Roper, 2025).

The reviewed features of various e-repository systems provided valuable insights into the essential components necessary for efficient data storage,

retrieval, and management in a research setting. These include functionalities such as user authentication, data categorization, advanced search tools, and secure data backup mechanisms. By integrating these features, the proposed E-Repository of Researches for the Camarines Norte Lowland Rainfed Research Station (CNLRRS) aims to enhance research accessibility, promote knowledge sharing, and improve data preservation within the institution. Ultimately, the study contributes to the digital transformation of CNLRRS by providing a sustainable and user-centered platform that supports research productivity and institutional data management efficiency.

Effectiveness of the E-Repository of Researches for Camarines Norte Lowland Rainfed Research Station (CNLRRS)

According to Agnihotri et al. E-Library, from the start of the twenty-first century, we have witnessed rapid information and Communication Technology development, so nowadays, with physical settings of the library, various online databases are available with large collections and are maintained by research institutions. A researcher can approach the particular e-library according to their needs as per procedure laid

down by the concerned library owner institution or organization.

The study investigated the availability and use of electronic databases by scientists in agricultural research institutes. Findings revealed that while electronic resources are available, their usage is hindered by limited awareness, inadequate training, and infrastructural challenges. The authors recommend capacity building, institutional support, and improved access policies to ensure maximum utilization of these e-resources (Magaji, 2024).

The study examined how undergraduate students of agriculture access library services for research. It highlighted the role of e-library tools such as TEEAL and online journals, recommending expanded access and improved training to strengthen academic research (Idiku et al., 2022).

This article highlighted how modern libraries can support agricultural scholarship by offering data management services, digital repositories, and open-access platforms. It concluded that libraries must expand beyond traditional roles to meet agricultural researchers' evolving needs (Van Camp & Hines, 2021).

Institutional adoption of e-libraries for agricultural education has shown positive results when integrated into teaching, learning, and extension activities. This study recommends aligning e-library services with curricula to maximize impact on research and farming practices (Rai & Pandey, 2022).

This article discussed how libraries can support AI-driven agriculture and food security in the Fourth Industrial Revolution. It emphasized the role of libraries in providing e-resources, managing datasets, and offering training for researchers and farmers. The authors call for capacity building and stronger partnerships to maximize digital opportunities (Adigun et al., 2024).

The modern innovative educational technologies projected in the twenty first century e-learning education, blended learning, distance learning, mobile learning (mlearning), virtual learning and e-library, provided opportunities for members of the academic communities such as professors, lecturers, instructors, teachers, researchers and students to acquire, develop and maintain core skills in essential digital knowledge without obstructions (Matthew et al., 2021).

The study findings indicated the primary motivations for utilizing e-resources were for assignments and research purposes. Among the e-resources, e-journals, e-books, and e-magazines were the most commonly accessed by the students. While the students held positive perceptions of the e-resources, the most significant challenge they faced was a lack of sufficient computers in the e-library. Based on the study's findings, it was recommended that efforts be made to encourage female students to utilize e-resources similar to their male counterparts (Mohammed et al., 2023).

E-library resources were the least subscribed support services among the students. In terms of social support services, interaction with tutors and interaction with fellow students came up as the main support services available to students, while counselling services were absent for these students. It was recommended that the support services that were inadequate, least used, and absent should be provided to enable students to complete their programmes of study successfully (Amponsah et al., 2021).

Libraries should subscribe to online libraries tailored for individuals with reading disabilities, like SugamyaPustakalaya. This e-library provides access to a

vast collection of resources for member libraries and their users. Introduce AI tools in university libraries to enhance the quality of services provided to users with reading disabilities. These tools can assist in personalized recommendations and improving accessibility features (Arunkumar, 2025).

In this research the focus is upon digital libraries and its importance for various sectors. E-Library or eGranthalaya is an example of it. With the help of these applications, students can take advantage of this facility by registering online. Resources available in the libraries of colleges and universities can be uploaded on it so that other colleges and universities can take advantage of them (Gurav, 2023).

In today's digital age, libraries have adopted technological innovation and provided facilities like digital repository, e-library, data management service, and research analysis tools, which are a revolutionary step towards innovation. Libraries foster a collaborative environment where exchange of ideas, dialogue and creative thinking are supported. Librarians act as catalysts of innovation, playing a vital role in directing users to appropriate resources and guiding them on the path of innovation (Prakashan, 2021).

The traditional library system is increasingly moving towards an e-digital library system. Because the traditional library system consumes a lot of valuable time, the e-library saves everyone's time. These e-resources are easily accessible via ICT tools (smartphones, laptops, tablets, and desktops). ~~In~~ During the COVID-19 pandemic, e-resources served as an essential medium for delivering information to students who were unable to attend regular classes or access the library (Prajapati, 2021).

E-libraries offer a wide array of resources, including e-books, academic journals, articles, multimedia content, and databases. This variety caters to diverse learning and research needs across different subjects and disciplines. Potentially reduce costs associated with maintaining physical collections and facilities. This can lead to more efficient use of resources and potentially lower costs for users in terms of access fees or subscription charges. By reducing the need for physical materials, e-libraries contribute to environmental sustainability by minimizing paper usage and transportation-related emissions (Vinayak, 2024).

University libraries now depend on both electronic and print media, based on network and physical

facilities. In this way, university libraries are able to offer their patrons remote services and access to the vast networked resources. The users of this type of library (e-library) require basic ICT and information searching skills, unlike in traditional libraries, where users are required to have only the ability to read (Unamma, 2020).

Library outreach services involve delivering information resources to users outside the traditional library building, ensuring that individuals who cannot physically visit the library still gain access to essential information. This includes initiatives such as bookmobiles, "library on the go," information buses, and elderly outreach projects. With the integration of modern information technology, these outreach services can bridge the gap between distant or underserved users and library facilities, enabling speedy and real-time information delivery and exchange. Such innovations highlight the transformative potential of library services in enhancing accessibility and inclusivity in the digital age (Johnson & Hill, 2022).

In conclusion, the studies reviewed show that there are already a lot of systems for managing research data. However, most of them are meant for general or academic

use. None of them are made just for research stations like CNLRRS. Hence, this study is important+ as it fills that gap by giving researchers an easy way to store, organize, and access their work online. This will help the institution share knowledge and make research management more efficient.

Synthesis of the Art

The literature reviewed here proves that digital knowledge systems and e-libraries have been essential facilitators of agricultural research and innovation between 2021 and 2025. Examples of existing platforms like the CABI Digital Library, TEEAL, and the USDA's Ag Data Commons show how digital centralized repositories can enhance access to information, increase research productivity, and furnish decision-support materials to researchers, farmers, and extension workers. Similarly, research in agricultural digitalization points to the growing incorporation of multimedia tools, data sets, and intelligent decision-support systems within digital repositories, going beyond the role of archives.

In spite of these strides, there are still gaps in localized accessibility, end-user training, sustainability, and data integration for production analysis at the institutional level. Although global and

national platforms offer general resources, a majority of farming research offices—even in localized settings—continue to fall short of having tailored e-libraries that deal specifically with their individualized data management and production monitoring requirements. Trustworthiness, governance, and usability are also considered persistent challenges that need to be resolved in order to achieve long-term uptake and usefulness.

This cutting-edge approach emphasizes the potential for the current study contribute by coming up with a design and implementation of a System. As opposed to generic platforms, this system will be a specialized e-library and knowledge center specific to the needs of CNLRRS, facilitating effective access to production data, optimized information management, and facilitation of research-driven decision-making. As such, the project pushes the art forward by filling in the gap between international e-library prototypes and specific agricultural research needs in localized settings, so that digital knowledge systems become both inclusive and contextually appropriate and sustainable.

Chapter 3

Methodology

The method for the study *E-Repository of Researches for Camarines Norte Lowland Rainfed Research Station (CNLRRS)* was presented in this chapter. This contains the software development methodology, project scheduling, sources of data, population of the study, data collection method, data analysis, and ethical considerations.

Software Development Methodology

A system development methodology was an organized strategy for planning, managing, and regulating the software development process. This study utilized the Agile methodology, specifically the Scrum framework, to develop the *E-Repository of Researches for Camarines Norte Lowland Rainfed Research Station (CNLRRS)*. This approach was selected to ensure that the final system was highly responsive to user needs through iterative development and continuous feedback from researchers and agricultural experts.

The Agile methodology was an iterative and incremental approach, organized into time-boxed development cycles known as sprints. Each sprint encompassed key activities including requirements

refinement, design adaptation, development, continuous feedback, testing, and deployment, resulting in a functional product increment that delivered value early and consistently.

The system's research and development were guided by the Agile methodology, which allowed constant adaptation to the evolving needs of CNLRRS. The team collaborated in iterative sprints to analyze research data management requirements, enhance system functionalities, and design a user-friendly repository platform. Regular feedback loops with stakeholders enabled quick adjustments during implementation, while continuous evaluation of usability influenced each successive development cycle. This method provided the prompt release of actionable, value-driven features for optimizing research data accessibility.

The Product Backlog was a prioritized list of all potential features, research projects, and enhancements for the CNLRRS Research Repository. It served as a constantly evolving blueprint based on feedback from researchers and stakeholders.

During the first requirements-gathering phase, the team conducted structured interviews and surveys with key stakeholders, including CNLRRS researchers, Daet-based

pineapple producers, and agricultural organizations such as the Provincial Agriculture Office (PAO).

The insights gained were used to identify and prioritize important system features such as file upload, user authentication, search and filter options, and research categorization. These were documented in the initial Product Backlog, ensuring that the system development plan was aligned with user needs.

The Sprint Backlog was a specific, time-bound group of work chosen from the Product Backlog for a single development sprint (usually 1-2 weeks). It demonstrated the team's genuine commitment to delivering a functional increment of the repository.

In this phase, the team selected key tasks for development, such as the login and registration module, advanced search filter, and secure file upload interface. Each member was assigned responsibilities, and clear sprint goals were set to ensure timely and organized progress. The team also held short daily stand-up meetings to monitor progress and resolve minor development issues.

Planning was the phase where the team defined what would be done, why it would be done, how it would be

done, and when it would be completed. This roadmap kept the project organized and achievable.

During this phase, the team conducted Sprint Planning meetings to decide which features to prioritize for each sprint. A timeline and task breakdown were created, the tools to be used (PHP, MySQL, HTML, CSS, and JavaScript) were determined, and the design mockups were finalized using Figma.

The planning process also included reviewing feedback from the adviser and CNLRRS staff to ensure that the development goals matched actual user requirements.

Implementation: This is where you actively follow the plan you created and put your ideas into action.

During the implementation, the Implementation phase is dedicated to developing and building the sprint backlog's features for the CNLRRS Research Repository, such as cultivation-focused search filters and secure user authentication. The team overcomes obstacles and stays on track by working together every day. By the end of the sprint, they have delivered a tested and working product update that makes pineapple production research more accessible.

Implementation: This was the phase where the team actively followed the plan and put ideas into action.

During the Implementation phase, the team developed and coded the system modules based on the sprint backlog. The database structure was created in MySQL, the front-end interface was developed, and the back-end functionalities, such as file upload, user authentication, and search filters, were integrated.

Daily coordination helped the team resolve technical issues quickly and maintain the project schedule. By the end of each sprint, a working and tested update of the system was produced for stakeholder review.

Review: This was the phase where the team carefully examined and evaluated the output and results of each sprint.

During this phase, the team presented the working system increments to CNLRRS staff and gathered feedback on usability, functionality, and design. The completed features were compared with the sprint goals to ensure that the outputs met project objectives. Any identified errors or improvement suggestions were recorded and prioritized for the next sprint cycle.

Retrospective: The Retrospective was the final phase of the Agile methodology and served as a reflection on the entire sprint process.

During this phase, the team evaluated what went well, what challenges were faced, and what improvements could be made for future sprints. The team revised system features based on client feedback, finalized the Data Management Information System flow, and refined the mock-ups for clarity and usability.

Finally, the team deployed the complete E-Repository system to CNLRRS for validation and initial implementation, ensuring that it met both the functional requirements and the expectations of the users.

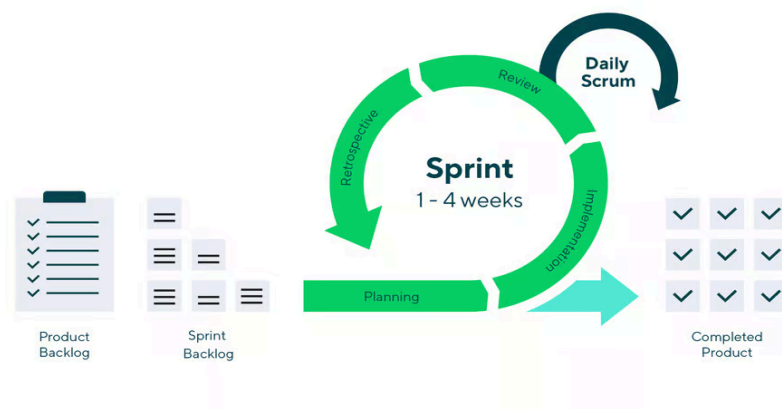


Figure 1. Agile Methodology Model

Schedule of Activities

The activities involved in the development of the Web Integrated Data Management Information System for Production Analysis of Camarines Norte Lowland Rainfed Research Station are presented in the figure below.

Figure 2 shows the Gantt Chart showing the duration of activities researchers must complete to accomplish the capstone project. They used the Gantt Chart as shown in Figure 2 to show the dates of the schedule of activities, which included the start and end of the activities, as well as the project activities summary. This visual representation helped to outline the timeline clearly.

The development of the E-Repository was guided by a six (6) phase process methodology designed to ensure quality and alignment with user needs. The process was developed in an orderly structure from Planning, Designing, Development, System Testing, Deployment, and concluded with the Feedback and Improvement phase, creating a cycle of ongoing improvement essential for the project's success. Planning: Finalization of Work Flow and System Features. This task is carried out by conducting group discussions and consultations with CNLRRS staff to identify the necessary system functions. The team analyzes current research management practices and determines the most efficient workflow that fits the organization's needs.

Information gathering and Workload Distribution. The team gathers information through interviews, document reviews, and analysis of existing repositories. After

data collection, tasks are distributed among members according to their skills to ensure a systematic and balanced workload.

Designing: Creation of Mockup and System Wireframe. This activity is done using design tools to visualize the system interface. The team drafts wireframes that represent the layout and user flow, which serve as a reference for developers during implementation.

Data Modelling. The process involves creating diagrams and defining database schemas that organize data entities, relationships, and attributes. It ensures that all research data can be efficiently stored, accessed, and managed.

Refinement based on feedback. The team gathers feedback from advisers and stakeholders after presenting the initial design. Suggestions and improvements are reviewed and integrated into the updated system design to enhance functionality and usability.

Development: Develop Core Features: Research Upload and Search. The development team codes and implements the system's primary functions, enabling users to upload, search, and access research files easily. This is done using programming languages and frameworks suited for web-based systems.

Coding of Frontend and Backend Functionalities. The frontend development focuses on creating the user interface, while the backend handles data processing and logic. Both are developed simultaneously to ensure system consistency and interaction between modules.

Integrate Backend with Frontend Systems. Integration is achieved by connecting APIs or database queries to the frontend interface. This allows dynamic data display and ensures that all user actions correspond accurately with stored information.

Improve UI, Performance, Fix Bugs. The team performs system optimization by testing the user interface, resolving performance issues, and fixing bugs detected during trial runs. This continuous improvement ensures system stability and user satisfaction.

System Testing: Unit Testing for each repository feature. All key repository functions, such as search, document upload, and user authentication, are thoroughly tested by the developers. Before the system gets unified, each component is evaluated separately to ensure reliable performance and operational effectiveness and error handling.

Integration Testing. The developers conduct integration testing to verify how linked modules such as

search, user authentication, and document upload systems—interact with one another. This guarantees system coherence and smooth data flow across all integrated components.

User Acceptance Testing (Researchers and CNLRRS Staff), The developers conduct User Acceptance Testing (UAT) with CNLRRS researchers and administrative staff to validate the system against real-world workflows and requirements. This last check guarantees the repository satisfies the actual needs and is prepared for deployment.

Collect Bug Reports and Usability Feedback. The Developers precisely gather and document all bug reports and usability feedback from UAT participants. Before the system's actual deployment, this process finds important problems and user experience improvements that are required.

Deployment: Preparation for Deployment. To get ready for the live launch, the developers complete server setup, data migration, and system configuration. This guarantees that the repository is prepared for operational use and that all technical requirements are fulfilled.

Deploy System Online. The fully-tested repository is deployed to a live web hosting platform by the developers. This signifies the platform's formal launch and opens the system to end users.

Migrate research data to production. In order to guarantee that the CNLRRS repository starts with an accurate and functioning digital collection for instant access by researchers and stakeholders, the developers upload all authenticated research materials and metadata to the live system.

Feedback and Improvement: Gather Feedback on Usability and Performance. The team collects structured feedback from CNLRRS staff and researchers regarding system usability and performance. This feedback is essential for identifying problem areas and improving the user experience as a whole.

Refinements Based on Feedback. Based on gathered user feedback, the developers make specific system enhancements. These improvements improve usability, fix performance problems, and guarantee that the repository better meets the specific needs of CNLRRS staff and researchers.

Conduct Surveys with Users. To get both quantitative and qualitative input, the team distributes standardized

surveys to CNLRRS researchers, employees, and stakeholders. For confirming the efficacy of the system and directing future improvements, this data offers important insights.

Final deployment of improved repository. The developers releases the system's final version, which includes all user input and performance enhancements. With this, the improved E-Repository is formally finished

and turned over to the Camarines Norte Lowland Rainfed

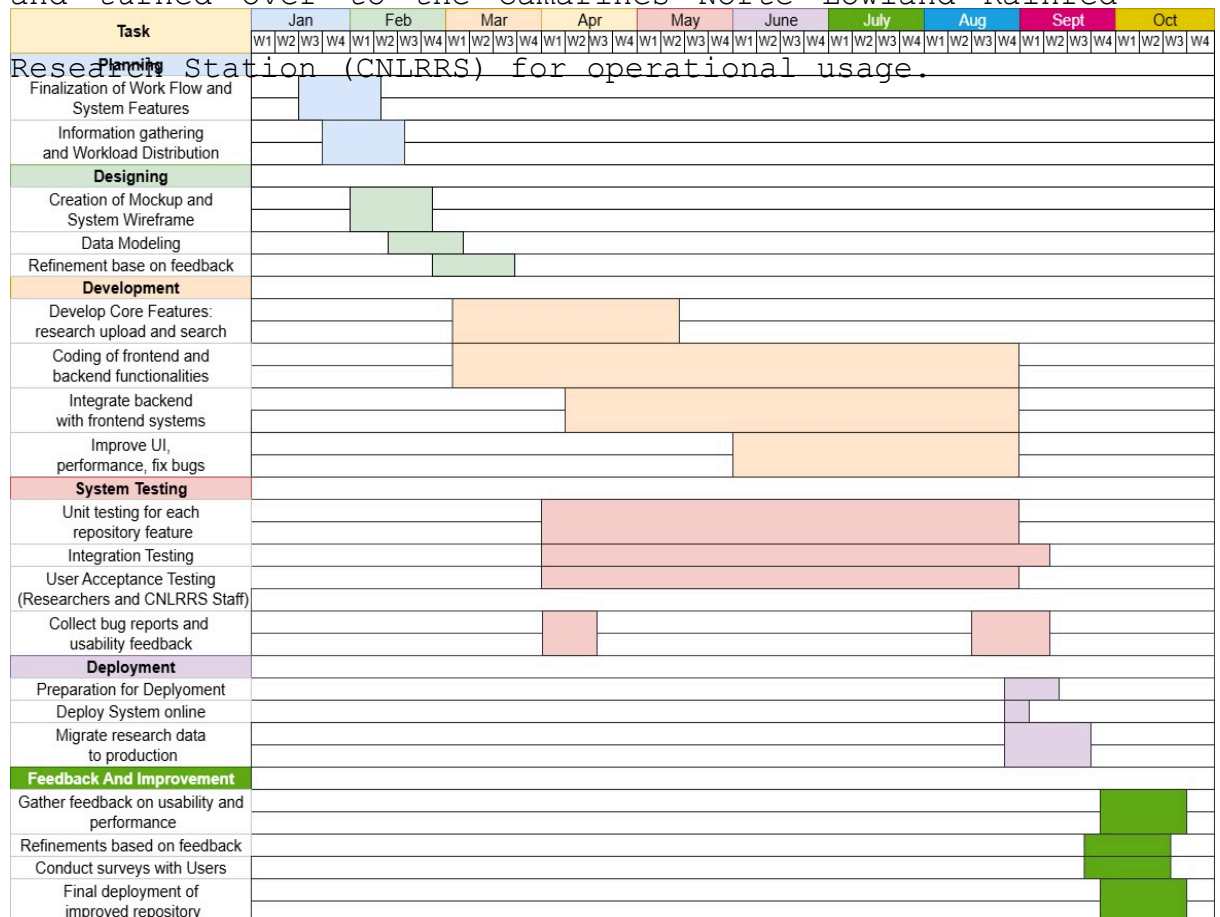


Figure 2. Schedule of Activities

Sources of Data

In this study, the researchers collected data from both primary and secondary sources. The primary data sources were responses from pineapple farmers, CNLRRS staff, and farm stakeholders engaged in crop production and marketing. They were gathered via structured interviews, focus group discussions, and questionnaires to gain information on their current practices, problems, and system requirements.

For Objective No. 3, which aimed to evaluate the system in terms of Functionality, Reliability, and Security, the data were obtained from system users and evaluators consisting of CNLRRS researchers, agricultural experts, and IT professionals. They participated in the system testing and evaluation process by completing assessment forms and usability checklists based on the ISO/IEC 25010 Software Quality Standards. Their feedback served as the primary basis for measuring the system's performance and overall quality.

The secondary data sources were in the form of existing research, related literature, and articles from the internet. These sources offered background information on farm data management systems, crop

marketing processes, and the adoption of technology among rural farm communities.

Population of the Study

The population of the study consisted of thirteen (13) respondents who were directly involved in the operations and management of the CNLRRS.

These respondents were selected because of their knowledge, experience, and engagement with the processes that the System aims to support. The researchers employed purposive sampling technique, wherein the participants were chosen in accordance with relevance to the study's purposes, that is, those having experience in data management, use of the system, or production analysis in CNLRRS.

Data Collection Method

To determine the main requirements and features, interviews were held with employees from various departments, such as production, system administration, and end-users of the existing manual process. The researchers also studied relevant literature and existing research on similar systems to gain further insight into best practices and design considerations.

Researchers also conducted a survey based on the evaluation of USE tools to identify the usability of the

existing process. The purpose of this survey was to determine the feedback provided by employees on the functionality and overall usefulness of the current data management tools. The result of the survey was used to solve research problem 3, which investigates the extent of usability of the existing system and how the usability can be enhanced.

To evaluate the efficiency of the new system in improving production analysis, qualitative and quantitative assessment approaches were applied. These were pre- and post-implementation surveys to measure improvement in system efficiency and production performance. The researchers also reviewed system logs and performance metrics to evaluate the effect on productivity.

Lastly, the researchers reviewed relevant research studies, industry reports, and documentation to validate the knowledge and ensure that the new system would be suitable for the unique needs of CNLRRS.

Data Analysis

The study employed a comprehensive data analysis approach, encompassing qualitative and quantitative analysis. The researchers blend qualitative and quantitative analysis for this study. Qualitative

analysis was used to investigate and understand CNLRRS personnel and pineapple farmers' experiences, problems, and expectations regarding their existing system. The method was highly appropriate in answering research questions 1 and 2. Quantitative analysis was used, however, to measure the efficacy of the system against research question 3. This allowed the researchers to translate subjective user comments into quantifiable data that could be subjected to statistical testing of the system's usability and effectiveness.

The researchers utilized the Likert scale, as shown in Table 1, to collect, analyze, and quantify the numerical data in determining the usability of mobile applications for e-learning. A Likert scale is a psychometric response scale primarily used in questionnaires to obtain participants' preferences or degree of agreement with a set of statements. Respondents rank quality from high to low or best to worst using five or seven levels (Mcleod, 2023).

Table 1
Likert Chart

Scale	Interval	Verbal Interpretation
5	4.21-5.00	Strongly Agree
4	3.41-4.20	Agree

3	2.61-3.40	<i>Neutral</i>
2	1.81-2.60	<i>Disagree</i>
1	1.00-1.80	<i>Strongly Disagree</i>

Statistical Treatment of Data

In this study, the data gathered from the 13 respondents were analyzed using descriptive statistics. Frequency counts and percentages were used to present the profile of the respondents and to summarize categorical responses such as demographics and system usage. This method makes it easier to see the distribution of responses across different categories.

For questions measured on a Likert scale, the weighted mean was applied to determine the overall assessment of the respondents regarding the effectiveness, usability, and acceptability of the proposed system. The weighted mean provides an average score that reflects the collective perception of the participants. To give further meaning to the results, the computed means were interpreted using a five-point scale with descriptive equivalents such as *Strongly Agree*, *Agree*, *Neutral*, *Disagree*, and *Strongly Disagree*.

Additionally, the standard deviation was used to measure the variability of responses. This shows whether

the respondents' opinions were consistent with each other or if they varied widely.

Formula for Weighted Mean:

$$\bar{X} = \frac{\Sigma(w.x)}{N}$$

Where:

$\bar{X}$ = weighted mean

Σ = summation symbol

W = weight assigned to each response

x = frequency of response

N = total number of respondents

Ethical Consideration

In addressing ethical issues and concerns in this study, the researchers strictly adhered to recognize ethical standards to ensure the research process was safe, genuine, and respectful of the participants' rights. The confidentiality, security, and privacy of every participant involved were protected at all times. The information and personal data gathered remained confidential and were used solely for the purpose of this research. Any acknowledgment of the work of other authors was properly credited and referenced following the APA citation style. Participation in this study was

voluntary, and all respondents provided their informed consent prior to data collection.

The researchers also ensured the responsible use of Artificial Intelligence (AI) tools during the conduct of the study. AI tools, such as ChatGPT and grammar checkers, were used only to assist in improving the clarity, grammar, and organization of written content. All AI-generated suggestions were critically reviewed, verified, and edited by the researchers to ensure accuracy and originality. No AI tools were used to fabricate data, generate false information, or manipulate results. This approach ensured transparency, integrity, and accountability in all stages of the research process. All findings derived from primary and secondary sources were objectively and accurately presented to maintain the credibility and reliability of the study

Chapter 4

RESULT AND DISCUSSION

This chapter presents the results and discussions of the researchers on the system that we are currently developing titled "E-Repository of Researches for Camarines Norte Lowland Rainfed" along with its analysis and interpretation of the data acquired through surveys and feedback to answer the statement of the problems

which are as follows: (1) What information requirements are needed to develop the E-Repository of Researches for Camarines Norte Lowland Rainfed Research Station (CNLRRS) (2) What specific system features of the E-Repository of Researches for Camarines Norte Lowland Rainfed Research Station (CNLRRS); (3) and Determine the level of effectiveness of the proposed study in terms of: a.) Functionality, b.) Reliability, and c.) Security.

Information Requirements Needed in the Development of the E-Repository of Researches for CNLRRS

Technical Requirements. The technical structure for the E-Repository was designed around a reliable, scalable, and secure technical foundation. This involved implementing a modern technology stack, including a React.js frontend for a responsive user interface and a Node.js backend with a MySQL database for efficient data management. These core technologies were utilized to ensure the system's performance, reliability, and maintainability, thereby creating a stable digital platform that guarantees effortless access and data reliability for all users.

Document and Data Files. A primary requirement for the development of the E-Repository was the integration of existing research documents from the CNLRRS. These

important documents, including completed research papers, technical reports, and experimental data, were used as important data and structural basis for the information framework of the repository. This ensured the system was designed to digitally preserve and systematically organize the station's research, thereby transforming divided research data into a centralized digital storage that is easily accessible.

Research Content Requirements. The development of the E-Repository required accurate research materials from CNLRRS, organized in a structured digital format. Important requirements included complete research documents such as PDF/DOC files containing full studies and technical reports, detailed research titles and author information for proper accreditation, informative abstracts and plan Summary for quick overviews, accurate publication and completion dates for order arrangement, and thorough documentation of research methodologies and findings. These core elements formed the repository's information foundation, enabling systematic digital preservation and organized access to the station's complete agricultural research portfolio.

Metadata & Categorization Requirements. The E-Repository system required a structured metadata

framework to ensure the accurate classification and discoverability of researchers from CNLRRS. Essential requirements included specific crop categorization to reflect the station's agricultural focus, detailed research themes for domain-specific filtering, and comprehensive keyword tagging. Additionally, accurate author and affiliation details were necessary for proper accreditation, while completion years and geographic scope specific to barangays or municipalities in Camarines Norte were required for chronological and locational organization.

User Management Requirements. The E-Repository required a role-based system to control access and protect content integrity. This involved defining three key user roles: Administrators for full system control, CNLRRS Researchers to submit and manage their own studies, and Public Users to browse and download approved research. A secure login system was also essential to authenticate these users and ensure the platform remained a trusted and organized resource.

Search & Retrieval Requirements. The repository required reliable search tools to ensure users could efficiently find relevant research. Essential features included a basic keyword search across titles and

abstracts, combined with advanced filters for specific criteria like crop type, research category, and publication year. This direct and insightful retrieval system was fundamental to transforming the stored research into an accessible and Useful library of information.

System Administration Requirements. The repository required a dedicated administrative panel to ensure its smooth operation and content quality. Key requirements included a content moderation system for reviewing and approving new research submissions, user account management tools for managing roles and permissions, and basic system monitoring to track repository health. These features were essential for maintaining the platform's integrity, security, and long-term stability.

Features of the e-Repository of Researches for CNLRRS

The system architecture and feature set of the E-Repository of Researches for the Camarines Norte Lowland Rainfed Research Station (CNLRRS) were carefully developed to address particular challenges of managing digital data in the field of agricultural research. The following core system elements were implemented to ensure reliability, usability, and scalability:

User Type-Based Permissions. The E-Repository implements User Type-Based Permissions to securely manage authorization for different user types. CNLRRS researchers receive benefits to submit and manage their research content, while administrators keep full control of the system. Public users are granted secure access to browse and download published materials. This structured approach guarantees that CNLRRS staff can work together and that knowledge is shared with the larger audience in an orderly way (as seen in Figure 3).

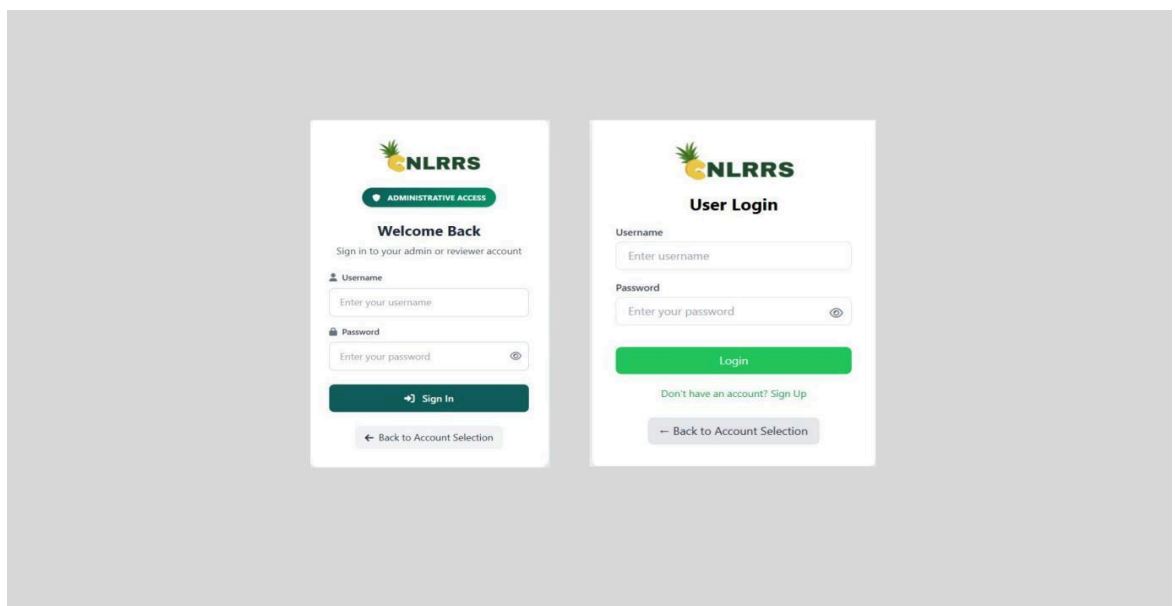


Figure 3. Role-Based Access

Advanced Search & Filtering. This core feature provides CNLRRS Staff and Researchers To conduct out specific searches over the whole research repository system. Users can locate specific studies using multiple

details including keywords, authors, publication years, and research methodologies. The convenient filtering system ensures precise and efficient discovery of relevant research studies, saving time and increasing the findings of research for the Camarines Norte lowland rainfed research station (as seen in Figure 4).

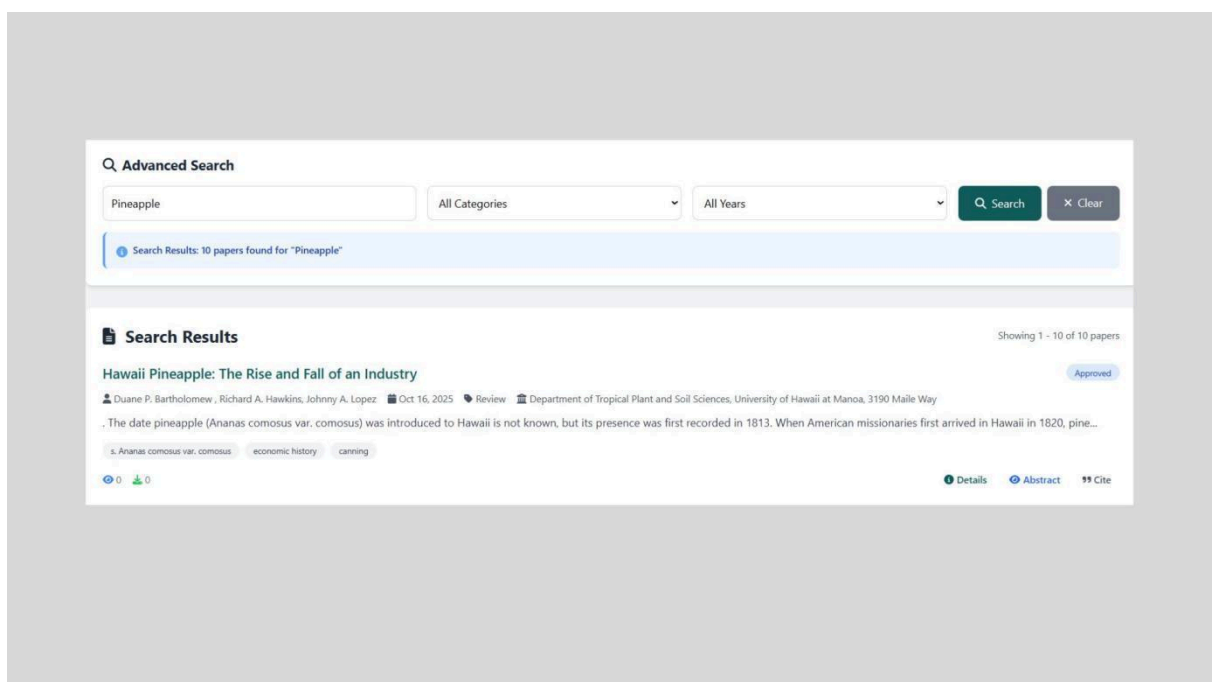


Figure 4. Advanced Search & Filtering

Research Submission & Workflow. This streamlined portal enables CNLRRS researchers to digitally submit their completed studies and datasets. Each submission enters an automated review and approval workflow to ensure content quality and validity. This structured process maintains academic standards while efficiently

expanding the repository's research collection (as seen in Figure 5).

The image shows a web form titled "1. Basic Information" and "4. Upload Research Paper". The "Basic Information" section includes a "Research Paper Title" field with a character count (0/500), a "Research Type" selection with five options: Experimental (Controlled studies), Observational (Field observations), Literature Review (Analysis of research), Case Study (Specific analysis), and Other (Mixed methods), and a "Keywords" field (5-8 keywords separated by commas). The "Upload Research Paper" section includes a "Research Paper File (PDF only, max 25MB)" field with a "Drop your PDF file here" area, a "Choose File" button, and a "or click to browse" link.

Figure 5. Research Submission & Workflow

Publication Process Tracker. This feature enables CNLRRS researchers and administrators to track the progress of research publications at each level, from submission and review by experts to repository admission and final approval. The tracker provides real-time visibility into the workflow, ensuring transparency and efficient management of CNLRRS's research pipeline (as seen in Figure 6).

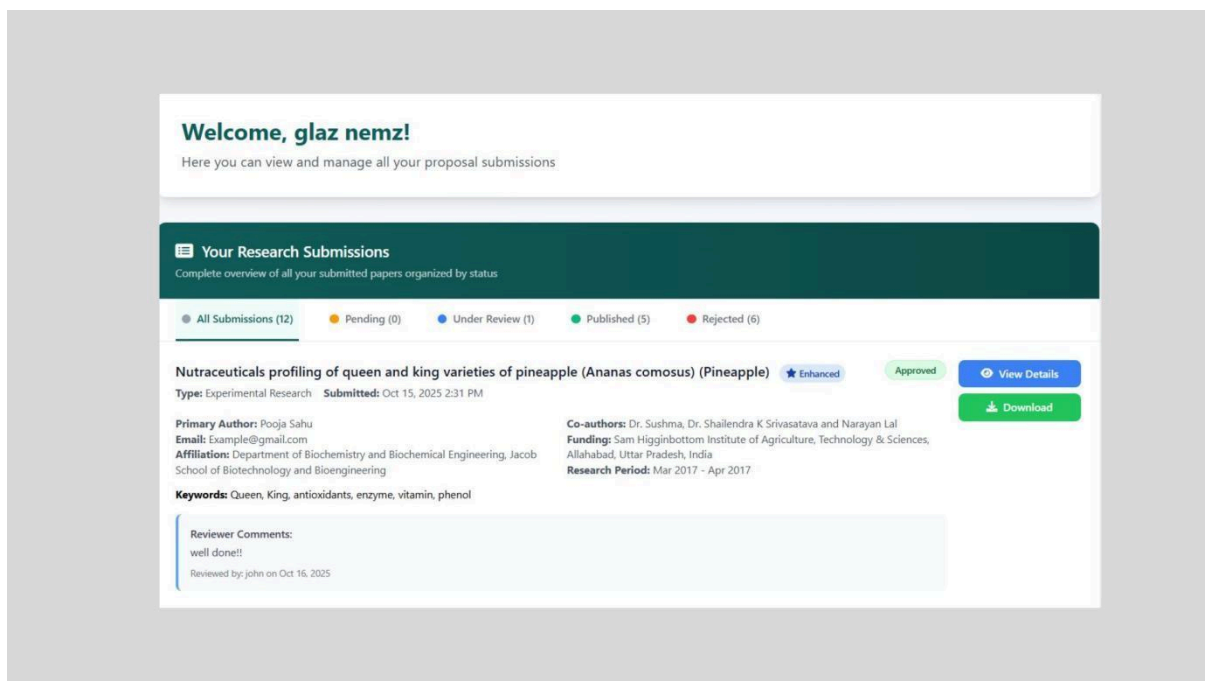


Figure 6. Upload Research Paper

Research Submission Status Updates. This feature provides automated, real-time notifications to users at every key stage of their research submission process. Authors will be notified as soon as their submission is successful, and they will later on receive progress updates when a conclusion is made regarding their paper is sent for review, approval or rejection, and whether any changes are needed. This ensures complete transparency and keeps researchers informed throughout the entire workflow (as seen in Figure 7).

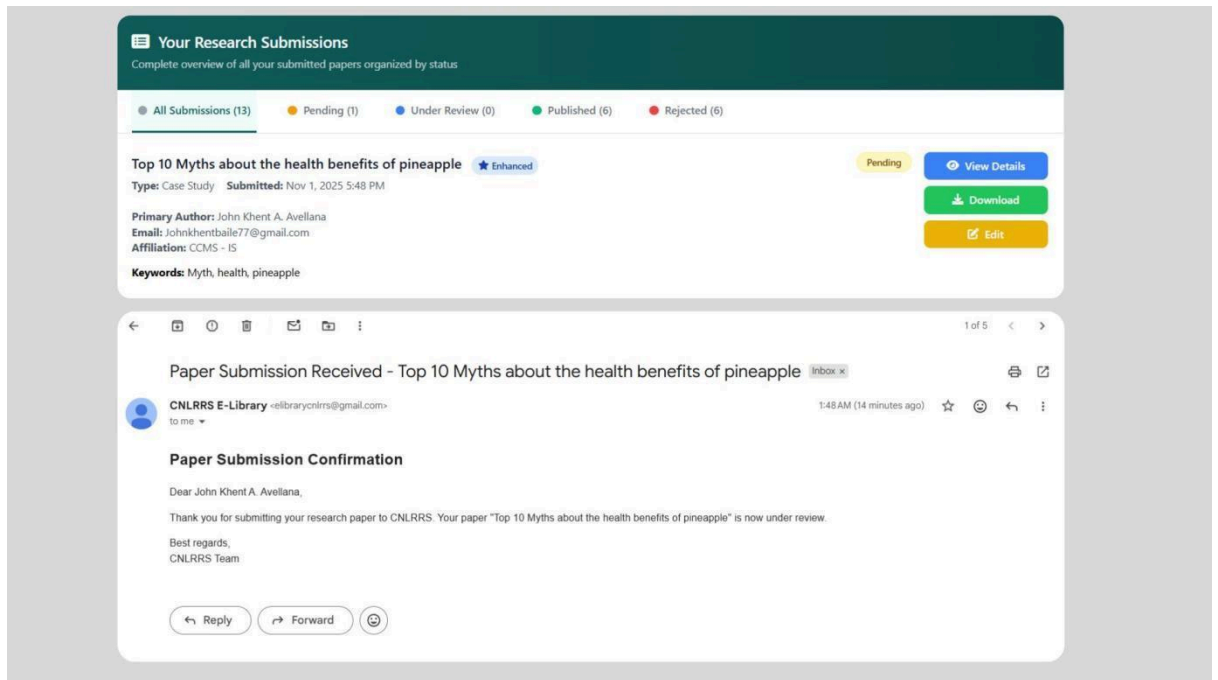


Figure 7. Terms and Submission Agreement

Level of Usability in Terms of Functionality, Performance Efficiency, Security, and Satisfaction

The proponents conducted an evaluation to gather user feedback in order to address the third research question of the study, which aimed at identifying the overall usability of the system.

Table 2
FUNCTIONALITY

Criteria	Respondents	Remarks
1. The system provides accurate and relevant information about research data.	4.46	Strongly Agree
2. The features of the system perform their intended functions effectively.	4.48	Strongly Agree
3. The search and filtering tools display the correct research results.	4.38	Strongly Agree
4. The upload and download features work properly.	4.44	Strongly Agree
5. The system meets the data management needs of CNLRRS.	4.46	Strongly Agree
Weighted Mean	4.44	Strongly Agree

As shown in Table 2, the evaluation results for the Functionality of the e-Repository for CNLRRS revealed highly positive feedback from respondents, with a

weighted mean of 4.44 (Strongly Agree). The highest mean score of 4.48 (Strongly Agree) was obtained for the criterion "The features of the system perform their intended functions effectively." This indicates that users found the system's functions efficient and reliable in supporting research-related tasks.

Overall, the lowest mean score of 4.38 (Strongly Agree) was recorded for "The search and filtering tools display the correct research results." Although slightly lower, this still reflects a strong agreement that the search feature effectively retrieves relevant research data. The results show that the e-Repository system is functional, dependable, and meets the data management needs of CNLRRS users.

Table 3
PERFORMANCE EFFICIENCY

Criteria	Respondents	Remarks
1. The system responds quickly to user action.	4.42	Strongly Agree
2. The loading time of pages and research data is acceptable.	4.44	Strongly Agree
3. The system can handle		Strongly Agree

multiple users at the same time without lagging.	4.33	
4. The system performs tasks efficiently and without unnecessary delay.	4.44	Strongly Agree
5. The overall performance of the system is stable and reliable.	4.5	Strongly Agree
Weighted Mean	4.43	Strongly Agree

As shown in Table 3, the evaluation results for the Performance Efficiency of the e-Repository for CNLRRS revealed very satisfactory outcomes, with a weighted mean of 4.43 (Strongly Agree). The highest mean score of 4.50 (Strongly Agree) was obtained for the criterion "The overall performance of the system is stable and reliable." This indicates that respondents found the system consistent and dependable in handling research-related operations.

Overall, the lowest mean score of 4.33 (Strongly Agree) was recorded for "The system can handle multiple users at the same time without lagging." Although relatively lower, this still shows strong agreement that

the system performs well even under multiple-user conditions. The findings demonstrate that the e-Repository operates efficiently, responds quickly, and maintains stable performance that meets user expectations.

Table 4
SECURITY

Criteria	Respondents	Remarks
1. The system ensures that only authorized users can access certain features.	4.35	Strongly Agree
2. Research data are protected from unauthorized modification or deletion.	4.38	Strongly Agree
3. The login and authentication process is secure.	4.5	Strongly Agree
4. User information is stored and handled confidentially .	4.5	Strongly Agree
5. I feel confident that the system protects research data	4.56	Strongly Agree

from breaches.		
Weighted Mean	4.46	Strongly Agree

As shown in Table 4, the evaluation results for the Security of the e-Repository for CNLRRS revealed highly favorable responses, with a weighted mean of 4.46 (Strongly Agree). The highest mean score of 4.56 (Strongly Agree) was obtained for the criterion "I feel confident that the system protects research data from breaches." This indicates that respondents trust the system's capability to safeguard sensitive research information against unauthorized access or potential security threats.

Overall, the lowest mean score of 4.35 (Strongly Agree) was recorded for "The system ensures that only authorized users can access certain features." Although slightly lower, it still reflects strong confidence that user permissions and access controls are effectively implemented. The findings show that the e-Repository maintains a high level of data protection, confidentiality, and secure user authentication, ensuring the reliability and safety of research data within CNLRRS.

Table 5
SATISFACTION

Criteria	Respondents	Remarks
1. I am satisfied with the performance of the e-Repository.	4.5	Strongly Agree
2. The system makes research data easier to manage and access.	4.58	Strongly Agree
3. I would recommend the system to other researchers or staff.	4.5	Strongly Agree
4. The interface of the system is simple and user-friendly.	4.52	Strongly Agree
5. The system contributes positively to the productivity of CNLRRS.	4.56	Strongly Agree
Weighted Mean	4.53	Strongly Agree

As shown in Table 5, the evaluation results for the Satisfaction of the e-Repository for CNLRRS revealed excellent feedback from respondents, with a weighted mean of 4.53 (Strongly Agree). The highest mean score of 4.58 (Strongly Agree) was obtained for the criterion "The system makes research data easier to manage and access."

This indicates that users found the e-Repository highly effective in simplifying research data handling and improving accessibility.

Overall, the lowest mean score of 4.50 (Strongly Agree) was recorded for both "I am satisfied with the performance of the e-Repository" and "I would recommend the system to other researchers or staff." Although slightly lower, these still reflect strong user approval and willingness to endorse the system to others. The results show that users are highly satisfied with the e-Repository's performance, usability, and contribution to the productivity and research efficiency of CNLRRS.

Chapter 5

CONCLUSIONS AND RECOMMENDATIONS

This chapter presented the findings, conclusions, and recommendations derived by the researchers that were relevant to the study.

Conclusions

The following are the conclusions drawn based on the findings of the study:

1. The E-Repository that was created functions effectively as a centralized, secure, and organized structured digital repository system. It solves the important problem of research documents at the Camarines Norte Lowland Rainfed Research Station (CNLRRS) that are spread out, hard to find, and quick to misuse. It combines years of useful agricultural research, experimental data, and technical reports into one organized platform. This makes it a reliable and long-lasting way to store data, manage knowledge, and share information easily between researchers, CNLRRS Staff, and other stakeholders.

2. The system's main features such as the advanced search tool with its filtering options, the simplistic categorization system that sorts by publication date, author, keyword and research themes that are relevant to

the Camarines Norte lowland, and the simple document upload portal—were thoroughly tested and received positive feedback from users to ensure that they fully met the information needs of its end-users. This set of features works together to solve the main problems of finding and accessing information. It greatly speeds up, improves, and makes the whole CNLRRS community's research retrieval and information sharing more efficient. This turns the repository from a simple digital archive into a useful research support tool.

3. The consistently high levels of acceptability recorded across key metrics—especially regarding its perceived usefulness in streamlining research workflows, its user-friendly design for individuals with varying technical skills, and its proven reliability in providing consistent access to critical data—offer clear evidence that the E-Repository is not just a functional system, but a robust, genuinely user-centric, and operationally mature solution. This validation confirms that the platform effectively achieves its primary goal of becoming an essential institutional asset, thus actively supporting and promoting evidence-based decision-making and advanced agricultural practices within the unique context of the Camarines Norte lowland rainfed area.

Recommendations

The following recommendations are proposed based on the conclusions:

1. The CNLRRS researchers and CNLRRS staff should officially adopt and use the established E-Repository as their main tool for storing, finding, and sharing local agricultural research.

2. The repository's files should be updated and updated continuously. To keep the resource up to date and complete, it is advised that CNLRRS management designate specialized staff to supervise the frequent submission and moderation of new research outputs, technical announcements, and information.

3. Establish a Data Management Policy: Develop and implement a formal data management policy that outlines the roles and responsibilities of researchers for uploading and updating their studies, ensuring the repository's content remains current and comprehensive.

4. For CNLRRS employees and stakeholders, a systematic training program and user guides should be created. This will provide consistent data entry procedures for long-term data quality and optimize the efficient use of all repository features.

5. Integration with outside agricultural research data and systems should be investigated in future development. To enhance the repository's content and offer a wider view of pineapple research in Camarines Norte Lowland Rainfed Research Station, it is advised to establish data-sharing protocols with regional and national agricultural research organizations.

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APPENDICES

Appendix A

Request Letter for Adviser

February 05, 2025

ALEXY GENE CASTILLO, DIT

College of Computing and Multimedia Studies
Camarines Norte State College

Sir:

Greetings!

We hope this letter finds you well. We are third-year BSIS students from Block A of the College of Computing and Multimedia Studies. As part of our IS-125 Capstone Project 2 course requirements, we humbly ask for assistance as our research adviser for our project entitled "e-Repository of Researches for Camarines Norte Lowland Rainfed Research Station (CNLRRS)."

In light of this, we kindly request for your expertise in order to aid and guide us during the development of our capstone project. We firmly believe that your knowledge and insights are invaluable and will play a pivotal role in the completion of this project.

We eagerly anticipate your affirmative response to our invitation. Thank you for committing your time to this request and we look forward to your positive response.

Sincerely yours,

John Kent A. Baile

BSIS 3A Student


Marielyn C. Ibana

BSIS 3A Student


Roniel McJain B. Dacillo

BSIS 3A Student


Mark Allen D. Raynes

BSIS 3A Student

Approved by:

ALEXY GENE CASTILLO, DIT

Appendix B

Letter for Pre-Final Defense

September 22, 2025

JOCELYN O. TORIO, DIT
Capstone Project 2 Professor
College of Computing and Multimedia Studies

Ma'am:

In partial fulfillment of the requirements for the Bachelor of Science in Information Systems program, we, the undersigned, have conducted a Capstone Project 2 study entitled "e-Repository of Researches for Camarines Norte Lowland Rainfed Research Station (CNLRRS)."

With the approved panel members:

- **EDGAR BRYAN B. NICART**
- **SPENCER S. SALUDES**

We would like to formally request the scheduling of our pre-final defense on **September 24, 2025**.

We look forward to your confirmation. Thank you very much for your guidance and support.

Sincerely,

John Khent A. Baile
BSIS 4A Student

Roniel McJain B. Dacillo
BSIS 4A Student

Marielyn C. Ibana
BSIS 4A Student

Mark Allen D. Raynes
BSIS 4A Student

Noted:

ALEXY GENE CASTILLO, DIT
Capstone Project 2 Adviser

Approved:

JOCELYN O. TORIO, DIT
Capstone Project 2 Professor

Appendix C

Request Letter for Evaluation

November 03, 2025

ENGR. EULA D. RADA
Science Research Specialist II
Research Unit Head

Ma'am:

Greetings!

We, the fourth-year students pursuing a Bachelor of Science in Information Systems a Camarines Norte State College, are reaching out to you about our Capstone Project, "e-Repository of Researches for Camarines Norte Lowland Rainfed Research Station (CNLRRS)."

In this regard, we kindly request your assistance and permission to conduct evaluation sessions on November 03, 2025. The purpose of these sessions is to complete the attached Usability Satisfaction Evaluation (USE) questionnaire. We assure you that all responses will remain confidential and will be used exclusively to evaluate the websites effectiveness. Your cooperation is greatly appreciated in helping us ensure the success of this evaluation.

We sincerely hope you will consider our request. We would greatly appreciate your support and look forward to your favorable response. Thank you very much for your time and consideration.

Sincerely yours,

John Khent A. Baile

BSIS 4A Student

Roniel McJain B. Dacillo

BSIS 4A Student

Noted by:

JOCELYN O. TORIO, DIT
Capstone Project 2 Instructor

ALEXT GENE CASTILLO, DIT
Capstone Project 2 Adviser

Approved by:

ENGR. EULA D. RADA
Research Unit Head

Marielllyn C. Ibana

BSIS 4A Student

Mark Allen D. Raynes

BSIS 4A Student

Appendix D

Invitation Letter for External Panel

November 26, 2025

ENGR. EULA D. RADA
Science Research Specialist II
Research Unit Head

Ma'am:

Greetings!

We hope this letter finds you well. We are fourth-year BSIS students from Block A of the College of Computing and Multimedia Studies. As part of our IS-125 Capstone Project 2 course requirements, we humbly ask for your assistance as our client for our project entitled **"e-Repository of Researches for Camarines Norte Lowland Rainfed Research Station (CNLRRS)."**

In light of this, we kindly request your expertise in order to aid and guide us during the presentation of our Capstone Project on **December 1 at 8:00 AM** onwards. We firmly believe that your knowledge and insights are invaluable and will play a pivotal role in the completion of this project.

We eagerly anticipate your affirmative response to our invitation. Thank you for committing your time to this request and we look forward to your positive response.

Sincerely yours,


John Khent A. Baile
BSIS 4A Student


Roniel McClain B. Dacillo
BSIS 4A Student

Noted by:
JOCELYN O. TORIO, DIT


Marielyn C. Ibana
BSIS 4A Student


Mark Allen D. Raynes
BSIS 4A Student

Approved by:
ENGR. EULA D. RADA

pp. 10/10/25
Received by:
[Signature]
11/26/2025

Appendix E

Letter for Final Defense

November 26, 2025

JOCELYN O. TORIO, DIT
Capstone Project 2 Professor
College of Computing and Multimedia Studies

Ma'am:

In partial fulfillment of the requirements for the Bachelor of Science in Information Systems program, the undersigned have conducted a Capstone Project 2 study entitled **"e-Repository of Researches for Camarines Norte Lowland Rainfed Research Station (CNLRRS)."**

With the approved panel members:

- **EDGAR BRYAN B. NICART, DIT**
- **SPENCER S. SALUDES**

We would like to formally request the scheduling of our final defense on **December 01, 2025.**

We look forward to your confirmation. Thank you very much for your guidance and support.

Sincerely,

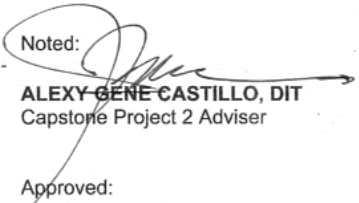

John Khent A. Baile
BSIS 4A Student


Roniel McClain B. Dacillo
BSIS 4A Student


Mariellyn C. Ibana
BSIS 4A Student


Mark Allen D. Raynes
BSIS 4A Student

Noted:


ALEXY GENE CASTILLO, DIT
Capstone Project 2 Adviser

Approved:

JOCELYN O. TORIO, DIT
Capstone Project 2 Professor

Appendix F

Sample Questionnaire

Name: _____ (optional) Date: _____
 Year/Course/Section _____

Instructions:

Kindly evaluate the **E-Repository of Researches for Camarines Norte Lowland Rainfed Research Station (CNLRRS)** by placing a check mark (☐) under the corresponding numerical rating with respect to its sub-criteria. Each rating is quantified by the following scale below.

RANGE	SCALE	DESCRIPTION
4.5-5.0	5	Strongly Agree
3.5-4.49	4	Agree
2.5-3.49	3	Moderately Agree
1.5-2.49	2	Slightly Agree
1.0-1.49	1	Disagree

FUNCTIONALITY	SCALE				
	1	2	3	4	5
The system provides accurate and relevant information about research data.					
The features of the system perform their intended functions effectively.					
The search and filtering tools display the correct research results.					
The upload and download features work properly.					
The system meets the data management needs of CNLRRS.					
PERFORMANCE EFFICIENCY	1	2	3	4	5
The system responds quickly to user actions.					
The loading time of pages and research data is acceptable.					
The system can handle multiple users at the same time without lagging.					
The system performs tasks efficiently and without unnecessary delay.					
The overall performance of the system is stable and reliable.					
SECURITY	1	2	3	4	5
The system ensures that only authorized users can access certain features.					
Research data are protected from unauthorized modification or deletion.					
The login and authentication process is secure.					
User information is stored and handled confidentially.					
I feel confident that the system protects research data from breaches.					
SATISFACTION	1	2	3	4	5
I am satisfied with the performance of the E-Repository.					
The system makes research data easier to manage and access.					
I would recommend the system to other researchers or staff.					
The interface of the system is simple and user-friendly.					
The system contributes positively to the productivity of CNLRRS.					

Thank you for your cooperation!

Appendix G

Source Code

approval/rejection:

```
<?php
session_start();

if (!isset($_SESSION['username']) ||
!in_array($_SESSION['role'], ['admin', 'reviewer'])) {
    header('Location: index.php');
    exit();
}

require_once 'connect.php';
require_once 'email_config.php';
require_once 'user_activity_logger.php';

if ($_SERVER['REQUEST_METHOD'] === 'POST' &&
isset($_POST['action'])) {
    header('Content-Type: application/json');

    try {
        $paper_id = (int)$_POST['paper_id'];
        $action = $_POST['action'];
        $reviewer_comments =
trim($_POST['reviewer_comments'] ?? '');
        $reviewer_name = $_SESSION['username'];

        if (!in_array($action, ['approve', 'reject',
'under_review'])) {
            throw new Exception('Invalid action');
        }

        $status_map = [
            'approve' => 'approved',
            'reject' => 'rejected',
            'under_review' => 'under_review'
        ];
        $new_status = $status_map[$action];
```

```

        $paper_sql = "SELECT ps.*, a.email as user_email,
a.name as author_full_name
                        FROM paper_submissions ps
                        LEFT JOIN accounts a ON ps.user_name
= a.username
                        WHERE ps.id = ?";
        $stmt = $conn->prepare($paper_sql);
        $stmt->bind_param('i', $paper_id);
        $stmt->execute();
        $paper = $stmt->get_result()->fetch_assoc();

        if (!$paper) {
            throw new Exception('Paper not found');
        }

        $user_email = $paper['user_email'] ??
$paper['author_email'] ?? null;
        if (!$user_email) {
            $email_search_sql = "SELECT email FROM
accounts WHERE username = ? OR name = ? LIMIT 1";
            $email_stmt =
$conn->prepare($email_search_sql);
            $email_stmt->bind_param('ss',
$paper['user_name'], $paper['author_name']);
            $email_stmt->execute();
            $email_result = $email_stmt->get_result();

            if ($email_row = $email_result->fetch_assoc())
{
                $user_email = $email_row['email'];
            }
        }

        if (!$user_email) {
            error_log("WARNING: No email found for paper
ID $paper_id, user: {$paper['user_name']}");
        }

        $conn->begin_transaction();

```

```

$update_sql = "UPDATE paper_submissions
               SET status = ?,
                   reviewer_comments = ?,
                   reviewed_by = ?,
                   review_date = NOW(),
                   updated_at = NOW()
               WHERE id = ?";
$stmt = $conn->prepare($update_sql);
$stmt->bind_param('sssi', $new_status,
                 $reviewer_comments, $reviewer_name, $paper_id);

if (!$stmt->execute()) {
    throw new Exception('Failed to update paper
status');
}

$notification_type = $new_status;
$notification_messages = [
    'approved' => 'Your paper has been approved
for publication.',
    'rejected' => 'Your paper requires revision.
Please review the feedback.',
    'under_review' => 'Your paper is now under
review by our team.'
];

$notif_sql = "INSERT INTO submission_notifications
              (paper_id, user_name,
notification_type, message, created_at)
              VALUES (?, ?, ?, ?, NOW())";
$stmt = $conn->prepare($notif_sql);
$notif_message =
$notification_messages[$new_status];
$stmt->bind_param('isss', $paper_id,
                 $paper['user_name'], $notification_type, $notif_message);
$stmt->execute();

$conn->commit();

$paper['reviewed_by'] = $reviewer_name;
$paper['review_date'] = date('F j, Y');

```

```

    $paper['reviewer_comments'] = $reviewer_comments;

    $email_sent = false;
    $email_error = null;

    if ($user_email) {
        try {
            // Attempt to send email
            $email_sent =
EmailService::sendPaperReviewNotification(
                $paper,
                $user_email,
                $new_status,
                $conn
            );

            if (!$email_sent) {
                $email_error = "Email send returned
false";
                error_log("Email send failed for paper
$paper_id to $user_email");
            }
        } catch (Exception $e) {
            $email_error = $e->getMessage();
            error_log("Exception sending email for
paper $paper_id: " . $email_error);
        }
    } else {
        $email_error = "No valid email address found";
        error_log("Cannot send email for paper
$paper_id: No email address");
    }

    $activity_msg = "Paper ID: $paper_id, Status:
$new_status, Reviewer: $reviewer_name";
    if ($email_sent) {
        $activity_msg .= ", Email: Sent to
$user_email";
    } else {
        $activity_msg .= ", Email: FAILED -
$email_error";
    }

```

```

    }
    logActivity('PAPER_REVIEWED', $activity_msg);

    echo json_encode([
        'success' => true,
        'message' => 'Paper review submitted
successfully',
        'new_status' => $new_status,
        'email_sent' => $email_sent,
        'email_info' => $email_sent
            ? "Email sent to $user_email"
            : "Warning: Email could not be sent -
$email_error. The review was saved and will be retried
automatically.",
        'email_to' => $user_email
    ]);

    } catch (Exception $e) {
        if ($conn) $conn->rollback();
        error_log("Review submission error: " .
        $e->getMessage());

        echo json_encode([
            'success' => false,
            'error' => $e->getMessage()
        ]);
    }
    exit();
}

$status_filter = $_GET['status'] ?? 'pending';
$search = $_GET['search'] ?? '';

$sql = "SELECT ps.*,
    COALESCE(view_count.views, 0) as total_views,
    COALESCE(download_count.downloads, 0) as
total_downloads
FROM paper_submissions ps
LEFT JOIN (
    SELECT paper_id, COUNT(*) as views
    FROM paper_metrics

```

```

        WHERE metric_type = 'view'
        GROUP BY paper_id
    ) view_count ON ps.id = view_count.paper_id
LEFT JOIN (
    SELECT paper_id, COUNT(*) as downloads
    FROM paper_metrics
    WHERE metric_type = 'download'
    GROUP BY paper_id
) download_count ON ps.id =
download_count.paper_id
WHERE 1=1";

if ($status_filter !== 'all') {
    $sql .= " AND ps.status = '" .
$conn->real_escape_string($status_filter) . "'";
}

if (!empty($search)) {
    $search_term = $conn->real_escape_string($search);
    $sql .= " AND (ps.paper_title LIKE '%$search_term%'
        OR ps.author_name LIKE '%$search_term%'
        OR ps.keywords LIKE '%$search_term%')";
}

$sql .= " ORDER BY ps.submission_date DESC";

$result = $conn->query($sql);
$papers = $result->fetch_all(MYSQLI_ASSOC);

$count_sql = "SELECT status, COUNT(*) as count FROM
paper_submissions GROUP BY status";
$count_result = $conn->query($count_sql);
$status_counts = [];
while ($row = $count_result->fetch_assoc()) {
    $status_counts[$row['status']] = $row['count'];
}

function getStatusBadge($status) {
    switch($status) {
        case 'pending':

```

```

        return 'bg-yellow-100 text-yellow-800 border
border-yellow-200';
        case 'under_review':
            return 'bg-blue-100 text-blue-800 border
border-blue-200';
        case 'approved':
            return 'bg-green-100 text-green-800 border
border-green-200';
        case 'rejected':
            return 'bg-red-100 text-red-800 border
border-red-200';
        default:
            return 'bg-gray-100 text-gray-800 border
border-gray-200';
    }
}
?>

```

Submission:

```

<?php
session_start();

if (!isset($_SESSION['username']) ||
empty($_SESSION['username'])) {
    if ($_SERVER["REQUEST_METHOD"] === "POST") {
        header('Content-Type: application/json');
        echo json_encode([
            'success' => false,
            'error' => 'Session expired. Please log in
again.',
            'redirect' => 'account.php'
        ]);
        exit();
    }
    header('Location: account.php');
    exit();
}

$current_username = $_SESSION['username'];

```

```

include 'connect.php';

if ($_SERVER["REQUEST_METHOD"] === "POST") {
    header('Content-Type: application/json');

    try {
        $username = $current_username; // Use the variable
we set earlier
        $required_fields = ['author_name', 'author_email',
'affiliateion', 'paper_title', 'abstract', 'keywords',
'research_type'];
        foreach ($required_fields as $field) {
            if (empty($_POST[$field])) {
                throw new Exception("Field '$field' is
required.");
            }
        }
        if (!isset($_FILES['paper_file']) ||
$_FILES['paper_file']['error'] !== UPLOAD_ERR_OK) {
            throw new Exception("Please upload a valid PDF
file.");
        }

        $file = $_FILES['paper_file'];
        $file_info = finfo_open(FILEINFO_MIME_TYPE);
        $mime_type = finfo_file($file_info,
$file['tmp_name']);
        finfo_close($file_info);

        if ($mime_type !== 'application/pdf') {
            throw new Exception("Only PDF files are
allowed.");
        }

        if ($file['size'] > 25 * 1024 * 1024) { // 25MB
            throw new Exception("File size must be less
than 25MB.");
        }

        if (!isset($_POST['terms_agreement'])) {

```

```

        throw new Exception("You must agree to the
terms and conditions.");
    }

    $author_name = trim($_POST['author_name']);
    $author_email =
filter_var(trim($_POST['author_email']),
FILTER_VALIDATE_EMAIL);
    if (!$author_email) {
        throw new Exception("Invalid email address
format.");
    }

    $affiliation = trim($_POST['affiliation']);
    $co_authors = isset($_POST['co_authors']) ?
trim($_POST['co_authors']) : null;
    $paper_title = trim($_POST['paper_title']);
    $abstract = trim($_POST['abstract']);
    $keywords = trim($_POST['keywords']);
    $methodology = isset($_POST['methodology']) ?
trim($_POST['methodology']) : null;
    $funding_source = isset($_POST['funding_source'])
? trim($_POST['funding_source']) : null;
    $research_start_date =
isset($_POST['research_start_date']) &&
!empty($_POST['research_start_date']) ?
$_POST['research_start_date'] : null;
    $research_end_date =
isset($_POST['research_end_date']) &&
!empty($_POST['research_end_date']) ?
$_POST['research_end_date'] : null;
    $ethics_approval =
isset($_POST['ethics_approval']) ?
trim($_POST['ethics_approval']) : null;
    $additional_comments =
isset($_POST['additional_comments']) ?
trim($_POST['additional_comments']) : null;
    $research_type = $_POST['research_type'];
    $terms_agreement =
isset($_POST['terms_agreement']) ? 1 : 0;

```

```

        $email_consent = isset($_POST['email_consent']) ?
1 : 0;
        $data_consent = isset($_POST['data_consent']) ? 1
: 0;

        if (strlen($abstract) < 100) {
            throw new Exception("Abstract must be at least
100 characters long.");
        }

        if (strlen($abstract) > 2000) {
            throw new Exception("Abstract cannot exceed
2000 characters.");
        }

        if ($research_start_date && $research_end_date) {
            $start = new DateTime($research_start_date);
            $end = new DateTime($research_end_date);
            if ($start > $end) {
                throw new Exception("Research start date
cannot be after end date.");
            }
        }

        $upload_dir = "uploads/papers/";
        if (!is_dir($upload_dir)) {
            if (!mkdir($upload_dir, 0755, true)) {
                throw new Exception("Failed to create
upload directory.");
            }
        }

        $file_extension = pathinfo($file['name'],
PATHINFO_EXTENSION);
        $unique_name = uniqid() . "_" . time() . "." .
$file_extension;
        $target_file = $upload_dir . $unique_name;

        if (!move_uploaded_file($file['tmp_name'],
$target_file)) {
            throw new Exception("Failed to upload file.");

```

```

    }

    $conn->begin_transaction();

    try {
$sql = "INSERT INTO paper_submissions
    (user_name, author_name, author_email, affiliation,
co_authors,
    paper_title, abstract, keywords, methodology,
funding_source,
    research_start_date, research_end_date,
ethics_approval, additional_comments,
    terms_agreement, email_consent, data_consent,
research_type, file_path,
    submission_date, status)
    VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?,
?, ?, ?, ?, NOW(), 'pending')";

$stmt = $conn->prepare($sql);
if (!$stmt) {
    throw new Exception("Database prepare error: " .
$conn->error);
}
$stmt->bind_param("sssssssssssssssiisss",
    $username, $author_name, $author_email, $affiliation,
$co_authors,
    $paper_title, $abstract, $keywords, $methodology,
$funding_source,
    $research_start_date, $research_end_date,
$ethics_approval, $additional_comments,
    $terms_agreement, $email_consent, $data_consent,
$research_type, $target_file
);

if (!$stmt->execute()) {
    throw new Exception("Database error: " .
$stmt->error);
}

$paper_id = $conn->insert_id;
$stmt->close();

```

```

        $notification_sql = "INSERT INTO
submission_notifications (paper_id, user_name,
notification_type, message)
                                VALUES (?, ?, 'submitted',
'Your paper has been successfully submitted and is
pending review.');"
        $notification_stmt =
$conn->prepare($notification_sql);
        if ($notification_stmt) {
            $notification_stmt->bind_param("is",
$paper_id, $username);
            $notification_stmt->execute();
            $notification_stmt->close();
        }
        $activity_sql = "INSERT INTO
user_activity_logs (user_id, username, activity_type,
activity_description, paper_id, ip_address)
                                SELECT id, username,
'submit_paper', 'Submitted research paper', ?, ?
                                FROM accounts WHERE username =
?";
        $activity_stmt =
$conn->prepare($activity_sql);
        if ($activity_stmt) {
            $ip_address = $_SERVER['REMOTE_ADDR'] ??
'unknown';
            $activity_stmt->bind_param("iss",
$paper_id, $ip_address, $username);
            $activity_stmt->execute();
            $activity_stmt->close();
        }
        $conn->commit();
        require_once 'email_config.php';

        $paperData = [
            'author_name' => $author_name,
            'paper_title' => $paper_title,
            'research_type' => $research_type,
            'user_name' => $username
        ];

```

```

$emailSent =
EmailService::sendPaperSubmissionNotification(
    $paperData,
    $author_email,
    $conn
);
if ($emailSent) {
    error_log("Submission confirmation email sent
successfully to: $author_email for paper ID: $paper_id");

    if (function_exists('logActivity')) {
        logActivity('EMAIL_SUBMISSION_SENT', "Email sent
to: $author_email, Paper ID: $paper_id");
    }
} else {
    error_log("FAILED to send submission confirmation
email to: $author_email for paper ID: $paper_id");

    if (function_exists('logError')) {
        logError("Email send failed for submission to:
$author_email", 'EMAIL_SEND_FAILED');
    }
}

echo json_encode([
    'success' => true,
    'message' => 'Paper submitted successfully! ' .
($emailSent ? 'Confirmation email has been sent to your
email address.' : 'However, the confirmation email could
not be sent. Please check your spam folder or contact the
administrator.'),
    'paper_id' => $paper_id,
    'redirect' => 'my_submissions.php?success=1'
]);

    } catch (Exception $e) {
        $conn->rollback();
        throw $e;
    }

    } catch (Exception $e) {

```

```

        if (isset($target_file) &&
file_exists($target_file)) {
            unlink($target_file);
        }

        echo json_encode([
            'success' => false,
            'error' => $e->getMessage()
        ]);
    }

    $conn->close();
    exit();
}

$user_info = null;
if (isset($_SESSION['username'])) {
    $user_sql = "SELECT name, email FROM accounts WHERE
username = ?";
    $stmt = $conn->prepare($user_sql);
    if ($stmt) {
        $stmt->bind_param("s", $_SESSION['username']);
        $stmt->execute();
        $result = $stmt->get_result();
        $user_info = $result->fetch_assoc();
        $stmt->close();
    }
}

$conn->close();
?>

```

E-library:

```

<?php
require_once 'connect.php';

session_start();

```

```

if (!isset($_SESSION['id']) ||
!isset($_SESSION['username']) ||
!isset($_SESSION['role'])) {
    header("Location: elibrary.php");
    exit();
}

$user_id = $_SESSION['id'];
$username = $_SESSION['username'];
$user_name = $_SESSION['name'];
$user_role = $_SESSION['role'];
$search_keyword = isset($_GET['search']) ?
$_GET['search'] : '';
$category_filter = isset($_GET['category']) ?
$_GET['category'] : '';
$year_filter = isset($_GET['year']) ? $_GET['year'] : '';
$page = isset($_GET['page']) ? (int)$_GET['page'] : 1;
$papers_per_page = 10;
$offset = ($page - 1) * $papers_per_page;
$where_conditions = ["status IN ('approved',
'published')"];
$params = [];
$params_types = '';

if (!empty($search_keyword)) {
    $where_conditions[] = "(paper_title LIKE ? OR keywords
LIKE ? OR abstract LIKE ? OR author_name LIKE ? OR
co_authors LIKE ?)";
    $search_term = "%$search_keyword%";
    $params[] = $search_term;
    $params[] = $search_term;
    $params[] = $search_term;
    $params[] = $search_term;
    $params[] = $search_term;
    $params_types .= 'sssss';
}

if (!empty($category_filter)) {
    $where_conditions[] = "research_type = ?";
    $params[] = $category_filter;
    $params_types .= 's';
}

```

```

}

if (!empty($year_filter)) {
    if ($year_filter === 'older') {
        $where_conditions[] = "YEAR(research_end_date) <=
2020";
    } else {
        $where_conditions[] = "YEAR(research_end_date) =
?";
        $params[] = $year_filter;
        $param_types .= 'i';
    }
}

$where_clause = implode(' AND ', $where_conditions);
$count_sql = "SELECT COUNT(*) as total FROM
paper_submissions WHERE $where_clause";
$count_stmt = $conn->prepare($count_sql);
if (!empty($params)) {
    $count_stmt->bind_param($param_types, ...$params);
}
$count_stmt->execute();
$total_papers =
$count_stmt->get_result()->fetch_assoc()['total'];
$total_pages = ceil($total_papers / $papers_per_page);

$sql = "SELECT ps.*,
            COALESCE(SUM(CASE WHEN pm.metric_type =
'view' THEN 1 ELSE 0 END), 0) as total_views,
            COALESCE(SUM(CASE WHEN pm.metric_type =
'download' THEN 1 ELSE 0 END), 0) as total_downloads
FROM paper_submissions ps
LEFT JOIN paper_metrics pm ON ps.id = pm.paper_id
WHERE $where_clause
GROUP BY ps.id
ORDER BY ps.research_end_date DESC
LIMIT ? OFFSET ?";

$stmt = $conn->prepare($sql);
$params[] = $papers_per_page;
$params[] = $offset;

```

```

$param_types .= 'ii';
$stmt->bind_param($param_types, ...$params);
$stmt->execute();
$papers = $stmt->get_result()->fetch_all(MYSQLI_ASSOC);

$featured_papers = [];
if (empty($search_keyword) && empty($category_filter) &&
empty($year_filter)) {
    $featured_sql = "SELECT ps.*,
                        COALESCE(SUM(CASE WHEN
pm.metric_type = 'view' THEN 1 ELSE 0 END), 0) as
total_views,
                        COALESCE(SUM(CASE WHEN
pm.metric_type = 'download' THEN 1 ELSE 0 END), 0) as
total_downloads
                        FROM paper_submissions ps
                        LEFT JOIN paper_metrics pm ON ps.id =
pm.paper_id
                        WHERE ps.status IN ('approved',
'published')
                        GROUP BY ps.id
                        ORDER BY ps.research_end_date DESC
                        LIMIT 3";
    $featured_result = $conn->query($featured_sql);
    $featured_papers =
    $featured_result->fetch_all(MYSQLI_ASSOC);
}

$stats_sql = "SELECT
    (SELECT COUNT(*) FROM paper_submissions WHERE status
IN ('approved', 'published')) as total_papers,
    (SELECT COUNT(DISTINCT author_name) FROM
paper_submissions WHERE status IN ('approved',
'published')) as total_researchers,
    (SELECT COUNT(DISTINCT research_type) FROM
paper_submissions WHERE status IN ('approved',
'published')) as research_categories,
    (SELECT COUNT(*) FROM paper_submissions WHERE status =
'under_review') as active_projects";
$stats_result = $conn->query($stats_sql);
$stats = $stats_result->fetch_assoc();

```

```

$category_sql = "SELECT research_type, COUNT(*) as count
                FROM paper_submissions
                WHERE status IN ('approved', 'published')
                GROUP BY research_type
                ORDER BY count DESC";
$category_result = $conn->query($category_sql);
$categories = $category_result->fetch_all(MYSQLI_ASSOC);
$user_submissions_sql = "SELECT COUNT(*) as count FROM
paper_submissions WHERE user_name = ?";
$user_submissions_stmt =
$conn->prepare($user_submissions_sql);
$user_submissions_stmt->bind_param('s', $username);
$user_submissions_stmt->execute();
$user_submissions_count =
$user_submissions_stmt->get_result()->fetch_assoc()['count'];
$is_searching = !empty($search_keyword) ||
!empty($category_filter) || !empty($year_filter);
?>

```